

COMP9312 — Assignment 2 Solutions

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Q1. Graph homomorphism and induced subgraph matching

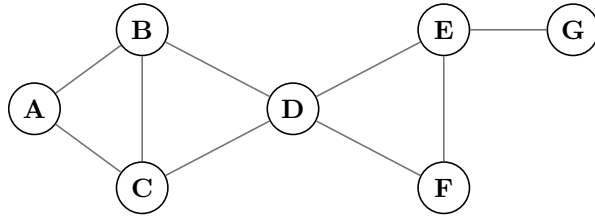


Figure 1: data graph G

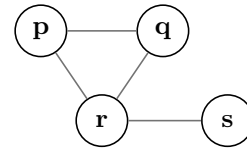


Figure 2: query graph P

$$E(G) = \{A-B, A-C, B-C, B-D, C-D, D-E, D-F, E-F, E-G\} \quad (9 \text{ edges});$$

$$E(P) = \{p-q, p-r, q-r, r-s\} \quad (4 \text{ edges}).$$

(a) A homomorphism from P to G

A homomorphism only has to map every *edge* of P to an edge of G , and distinct vertices of P may collapse onto one vertex of G — but only *non-adjacent* ones: G is simple, so an edge of P whose two endpoints collapsed would need a self-loop in G . Hence p, q, r , being pairwise adjacent, must land on three *distinct* vertices spanning a triangle of G , while the pendant s is free to reuse the image of p or q . Sending the triangle to B, C, D leaves r 's image D with the spare neighbour E for s , so nothing needs to collapse at all:

$$p \mapsto B, \quad q \mapsto C, \quad r \mapsto D, \quad s \mapsto E.$$

Edge of P	$p-q$	$p-r$	$q-r$	$r-s$
Image in G	$B-C \checkmark$	$B-D \checkmark$	$C-D \checkmark$	$D-E \checkmark$

Answer. A homomorphism from P to G exists, e.g. $p \mapsto B, q \mapsto C, r \mapsto D, s \mapsto E$ (all four edges of P land on edges of G).

(b) All induced subgraphs of G isomorphic to P

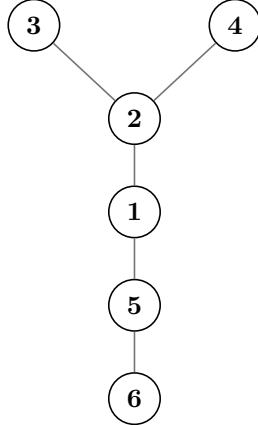
P is a triangle plus one pendant vertex (4 vertices, 4 edges). A vertex set $S \subseteq V(G)$ with $|S| = 4$ induces a copy of P **iff** three of its vertices form a triangle and the fourth sends *exactly one* edge into that triangle. An induced subgraph keeps *every* edge of G inside S , so two or three such edges would give 5 or 6 edges rather than 4, whereas zero edges would leave the fourth vertex isolated even though P is connected. Conversely any induced copy of P must contain a triangle, since P does, and its remaining vertex is P 's degree-1 pendant — so enumerating G 's triangles is exhaustive.

G has exactly three triangles: $\{A, B, C\}$, $\{B, C, D\}$, $\{D, E, F\}$. Extending each by a fourth vertex:

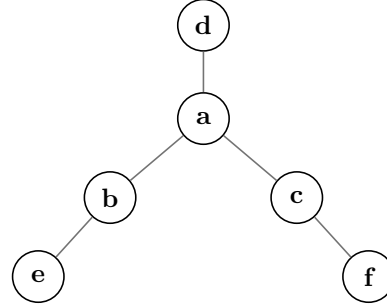
Triangle	4th vertex	Edges into the triangle	Induced copy of P ?
$\{A, B, C\}$	D	$D-B, D-C$ (2)	no
$\{A, B, C\}$	E	— (0)	no
$\{A, B, C\}$	F	— (0)	no
$\{A, B, C\}$	G	— (0)	no
$\{B, C, D\}$	A	$A-B, A-C$ (2)	no
$\{B, C, D\}$	E	$E-D$ (1)	yes
$\{B, C, D\}$	F	$F-D$ (1)	yes
$\{B, C, D\}$	G	— (0)	no
$\{D, E, F\}$	A	— (0)	no
$\{D, E, F\}$	B	$B-D$ (1)	yes
$\{D, E, F\}$	C	$C-D$ (1)	yes
$\{D, E, F\}$	G	$G-E$ (1)	yes

Answer. Exactly five vertex sets induce a subgraph isomorphic to P : $\{B, C, D, E\}$, $\{B, C, D, F\}$, $\{B, D, E, F\}$, $\{C, D, E, F\}$, $\{D, E, F, G\}$.

Q2. Weisfeiler-Lehman color refinement



G : a tree on 6 vertices



G' : a tree on 6 vertices

$$E(G) = \{1-2, 1-5, 2-3, 2-4, 5-6\}; \quad E(G') = \{a-b, a-c, a-d, b-e, c-f\} \quad (5 \text{ edges each}).$$

Both are trees on 6 vertices with the same degree sequence $[3, 2, 2, 1, 1, 1]$, so degrees alone cannot tell them apart.

Convention. A vertex's tuple is (own colour, sorted multiset of neighbour colours). Each round, the distinct tuples occurring in *either* graph are sorted lexicographically and receive the IDs $0, 1, 2, \dots$. Restarting the IDs at 0 in every round is what “starting from 0” prescribes. Continuing the numbering across rounds instead would only shift each round's IDs by a constant, padding the vectors with zero entries for the colours of earlier rounds; since $\phi_t(G)$ and $\phi_t(G')$ are always indexed the same way, every inner product — and hence the kernel — is unchanged either way.

Round 0

Every vertex has colour 0, so there is one colour and $\phi_0(G) = \phi_0(G') = [6]$.

Round 1

Lexicographic order of the tuples: $(0, [0]) < (0, [0, 0]) < (0, [0, 0, 0])$. (Comparing the colour first and then the neighbour list, or flattening each tuple into one sequence, give the same order in both rounds, so the answer does not depend on which reading is taken.)

ID	Tuple	Meaning
0	$(0, [0])$	degree-1 vertices
1	$(0, [0, 0])$	degree-2 vertices
2	$(0, [0, 0, 0])$	degree-3 vertices

G	Tuple	New colour	G'	Tuple	New colour
1	$(0, [0, 0])$	1	a	$(0, [0, 0, 0])$	2
2	$(0, [0, 0, 0])$	2	b	$(0, [0, 0])$	1
3	$(0, [0])$	0	c	$(0, [0, 0])$	1
4	$(0, [0])$	0	d	$(0, [0])$	0
5	$(0, [0, 0])$	1	e	$(0, [0])$	0
6	$(0, [0])$	0	f	$(0, [0])$	0

$$\phi_1(G) = [3, 2, 1], \quad \phi_1(G') = [3, 2, 1]$$

still identical — round 1 only counts degrees.

Round 2

Tuples are built from the round-1 colours above.

G	Colour	Neighbours' colours	Tuple	New colour
1	1	2(2), 5(1)	$(1, [1, 2])$	4
2	2	1(1), 3(0), 4(0)	$(2, [0, 0, 1])$	5
3	0	2(2)	$(0, [2])$	1
4	0	2(2)	$(0, [2])$	1
5	1	1(1), 6(0)	$(1, [0, 1])$	2
6	0	5(1)	$(0, [1])$	0

G'	Colour	Neighbours' colours	Tuple	New colour
a	2	$b(1), c(1), d(0)$	$(2, [0, 1, 1])$	6
b	1	$a(2), e(0)$	$(1, [0, 2])$	3
c	1	$a(2), f(0)$	$(1, [0, 2])$	3
d	0	$a(2)$	$(0, [2])$	1
e	0	$b(1)$	$(0, [1])$	0
f	0	$c(1)$	$(0, [1])$	0

Seven distinct tuples occur across the two graphs; sorted lexicographically they receive the IDs 0–6:

ID	Tuple	Count in G	Count in G'
0	$(0, [1])$	1	2
1	$(0, [2])$	2	1
2	$(1, [0, 1])$	1	0
3	$(1, [0, 2])$	0	2
4	$(1, [1, 2])$	1	0
5	$(2, [0, 0, 1])$	1	0
6	$(2, [0, 1, 1])$	0	1
total		6	6

$$\phi_2(G) = [1, 2, 1, 0, 1, 1, 0], \quad \phi_2(G') = [2, 1, 0, 2, 0, 0, 1]$$

Round 2 is the first round that separates the two trees: in G the degree-3 vertex sees two leaves and one degree-2 vertex, whereas in G' it sees one leaf and two degree-2 vertices.

(b) The two-round WL kernel

t	$\phi_t(G)$	$\phi_t(G')$	$\phi_t(G)^\top \phi_t(G')$
0	[6]	[6]	$6 \cdot 6 = 36$
1	[3, 2, 1]	[3, 2, 1]	$9 + 4 + 1 = 14$
2	[1, 2, 1, 0, 1, 1, 0]	[2, 1, 0, 2, 0, 0, 1]	$1 \cdot 2 + 2 \cdot 1 = 4$

Answer. $K_{\text{WL}}(G, G') = 36 + 14 + 4 = \mathbf{54}$.

Q3. Reachability indexing

(a) Minimal 2-hop labelling

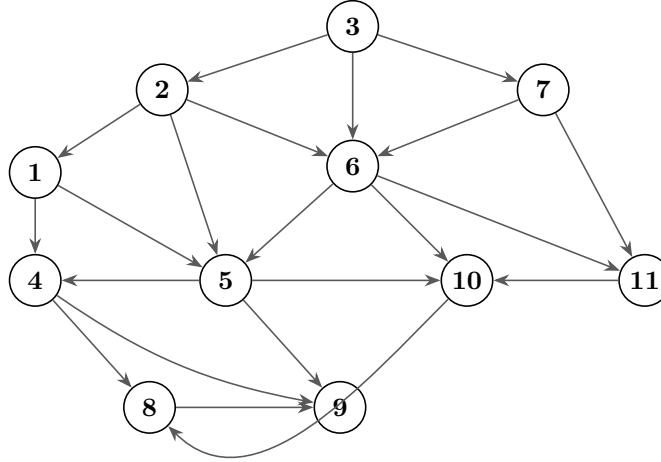


Figure 4: a DAG for minimal 2-hop labelling

Edges (21): $1 \rightarrow 4, 1 \rightarrow 5, 2 \rightarrow 1, 2 \rightarrow 5, 2 \rightarrow 6, 3 \rightarrow 2, 3 \rightarrow 6, 3 \rightarrow 7, 4 \rightarrow 8, 4 \rightarrow 9, 5 \rightarrow 4, 5 \rightarrow 9, 5 \rightarrow 10, 6 \rightarrow 5, 6 \rightarrow 10, 6 \rightarrow 11, 7 \rightarrow 6, 7 \rightarrow 11, 8 \rightarrow 9, 10 \rightarrow 8, 11 \rightarrow 10$.

Convention. $L_{\text{out}}(u)$ holds hubs reachable *from* u , $L_{\text{in}}(v)$ holds hubs that *reach* v , and

$$u \rightsquigarrow v \iff L_{\text{out}}(u) \cap L_{\text{in}}(v) \neq \emptyset.$$

Every vertex ends up in its own two labels. This is not a convention chosen by hand — it follows from the algorithm below. Each of hub k 's two searches starts *at* k , and the pruning test there is whether $L_{\text{out}}(k) \cap L_{\text{in}}(k)$ is already non-empty. In a DAG that intersection is necessarily empty before k labels itself: a common hub $h \neq k$ would give $k \rightsquigarrow h$ and $h \rightsquigarrow k$, i.e. a cycle. So k is never pruned at itself and always adds $k \in L_{\text{out}}(k)$, $k \in L_{\text{in}}(k)$. This is also exactly what makes the reflexive query $v \rightsquigarrow v$ answerable, since v is the only possible hub for it.

Processing order. Total degree = in-degree + out-degree, decreasing, ties broken by smaller vertex ID:

v	1	2	3	4	5	6	7	8	9	10	11
in-degree	1	1	0	2	3	3	1	2	3	3	2
out-degree	2	3	3	2	3	3	2	1	0	1	1
total	3	4	3	4	6	6	3	3	3	4	3

Order: 5, 6, 2, 4, 10, 1, 3, 7, 8, 9, 11

Algorithm. Take the hubs k in that order. For each k run a *backward* BFS (over reversed edges) and a *forward* BFS. On reaching a vertex x , first test whether the labels built *so far* already prove the corresponding reachability — $L_{\text{out}}(x) \cap L_{\text{in}}(k) \neq \emptyset$ backward, $L_{\text{out}}(k) \cap L_{\text{in}}(x) \neq \emptyset$ forward. If they do, **prune**: add nothing and do not expand past x . This is sound because the hub h that certifies the pair was processed *earlier* and lies on the path from k through x to everything beyond it, so each of those pairs is already answered by the labels built so far — though not necessarily through h itself. (At hub 2, for instance, the forward BFS prunes at 6 via hub 6, yet the pair

2 $\rightsquigarrow$ 4 behind it is certified by hub 5.) Otherwise add k to $L_{\text{out}}(x)$ (backward) or $L_{\text{in}}(x)$ (forward) and keep expanding. The result is a *minimal* 2-hop cover: no single entry can be deleted without breaking some reachable pair.

Round-by-round trace ($+v$ = hub added at v , $\times v$ = pruned at v):

Hub k	Backward BFS $\rightarrow L_{\text{out}}$	Forward BFS $\rightarrow L_{\text{in}}$
5	+1, +2, +3, +5, +6, +7	+4, +5, +8, +9, +10
6	+2, +3, +6, +7	+6, +11; $\times 5, \times 10$
2	+2, +3	+1, +2; $\times 4, \times 5, \times 6$
4	+4; $\times 1, \times 5$	+4, +8, +9
10	+10, +11; $\times 5, \times 6, \times 7$	+8, +9, +10
1	+1; $\times 2$	+1; $\times 4, \times 5$
3	+3	+3, +7; $\times 2, \times 6, \times 11$
7	+7; $\times 3$	+7; $\times 6, \times 11$
8	+8; $\times 4, \times 10$	+8, +9
9	+9; $\times 4, \times 5, \times 8$	+9
11	+11; $\times 6, \times 7$	+11; $\times 10$

Two examples of the pruning. At hub 6 the forward BFS stops at 5 because $5 \in L_{\text{out}}(6)$ and $5 \in L_{\text{in}}(5)$ already certify $6 \rightsquigarrow 5$; everything behind 5 (namely 4, 8, 9, 10) happens to be covered through hub 5 as well — this is one of the cases where the certifying hub does cover the whole tail — so the search need not go on. At hub 8 the backward BFS stops at 4 and at 10, whose labels $L_{\text{out}}(4) = \{4\}$ and $L_{\text{out}}(10) = \{10\}$ already meet $L_{\text{in}}(8) = \{4, 5, 10\}$.

Final labels.

v	$L_{\text{in}}(v)$	$L_{\text{out}}(v)$
1	$\{1, 2\}$	$\{1, 5\}$
2	$\{2\}$	$\{2, 5, 6\}$
3	$\{3\}$	$\{2, 3, 5, 6\}$
4	$\{4, 5\}$	$\{4\}$
5	$\{5\}$	$\{5\}$
6	$\{6\}$	$\{5, 6\}$
7	$\{3, 7\}$	$\{5, 6, 7\}$
8	$\{4, 5, 8, 10\}$	$\{8\}$
9	$\{4, 5, 8, 9, 10\}$	$\{9\}$
10	$\{5, 10\}$	$\{10\}$
11	$\{6, 11\}$	$\{10, 11\}$
total	23 entries	21 entries

Sanity checks: $3 \rightsquigarrow 9$ because $L_{\text{out}}(3) \cap L_{\text{in}}(9) = \{5\}$ (indeed $3 \rightarrow 2 \rightarrow 5 \rightarrow 9$); $1 \not\rightsquigarrow 11$ because $L_{\text{out}}(1) \cap L_{\text{in}}(11) = \{1, 5\} \cap \{6, 11\} = \emptyset$, and everything reachable from 1 does stay inside

$\{4, 5, 8, 9, 10\}$.

Answer. The minimal 2-hop labels are the 44 entries tabulated above (23 in L_{in} , 21 in L_{out}), obtained by pruned BFS with the hub order 5, 6, 2, 4, 10, 1, 3, 7, 8, 9, 11.

(b) Comparing two spanning trees

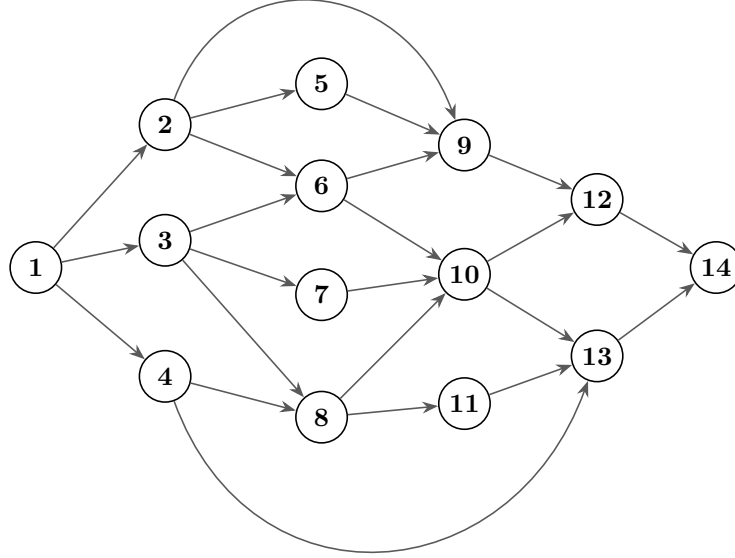


Figure 5: the DAG G (14 vertices, 23 edges)

Edges (23; vertex 1 is the only source): $1 \rightarrow 2, 1 \rightarrow 3, 1 \rightarrow 4, 2 \rightarrow 5, 2 \rightarrow 6, 2 \rightarrow 9, 3 \rightarrow 6, 3 \rightarrow 7, 3 \rightarrow 8, 4 \rightarrow 8, 4 \rightarrow 13, 5 \rightarrow 9, 6 \rightarrow 9, 6 \rightarrow 10, 7 \rightarrow 10, 8 \rightarrow 10, 8 \rightarrow 11, 9 \rightarrow 12, 10 \rightarrow 12, 10 \rightarrow 13, 11 \rightarrow 13, 12 \rightarrow 14, 13 \rightarrow 14$.

Algorithm. Number the tree vertices by a **left-to-right post-order** traversal. Each vertex v receives the interval

$$I(v) = [\min\{\text{post-numbers in } v\text{'s subtree}\}, \text{post}(v)],$$

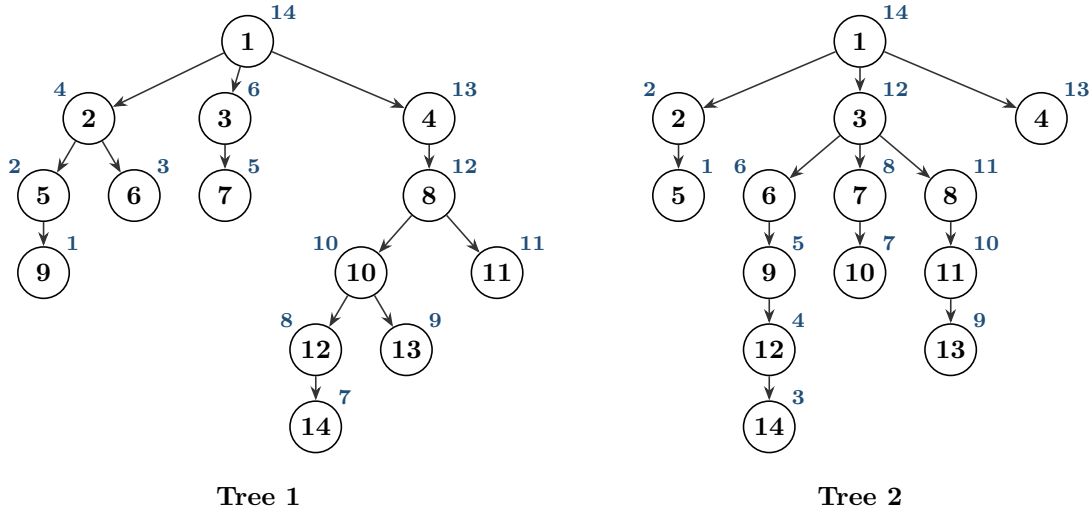
which covers exactly v 's tree descendants. Then, in **reverse topological order**,

$$L(u) = I(u) \cup \bigcup_{u \rightarrow w \in E} L(w),$$

deleting from $L(u)$ any interval *contained* in another interval of the same label — that is the redundancy the algorithm removes. A query is then $u \rightsquigarrow v \iff \text{post}(v)$ lies in some interval of $L(u)$.

Adjacent or partially overlapping intervals are *not* merged. Merging them would in fact stay correct, since post-numbers are consecutive integers and e.g. $[1, 2]$ together with $[3, 6]$ covers exactly $[1, 6]$; it is simply a further compression that the tree-cover algorithm does not perform. It would not affect the comparison either — merged, the two labellings below need 25 and 27 intervals instead of 26 and 31, so Tree 1 still wins.

Both spanning trees are rooted at vertex 1, the only source of the DAG. The post-order number of each vertex is printed in blue beside it.



Children, left to right —

Tree 1: 1: 2, 3, 4; 2: 5, 6; 3: 7; 4: 8; 5: 9; 8: 10, 11; 10: 12, 13; 12: 14.

Tree 2: 1: 2, 3, 4; 2: 5; 3: 6, 7, 8; 6: 9; 7: 10; 8: 11; 9: 12; 11: 13; 12: 14.

Tree 1. Post-order: 9, 5, 6, 2, 7, 3, 14, 12, 13, 10, 11, 8, 4, 1.

v	post	$I(v)$	Final $L(v)$	$ L(v) $
1	14	$[1, 14]$	$[1, 14]$	1
2	4	$[1, 4]$	$[1, 4], [7, 10]$	2
3	6	$[5, 6]$	$[1, 1], [3, 3], [5, 6], [7, 12]$	4
4	13	$[7, 13]$	$[7, 13]$	1
5	2	$[1, 2]$	$[1, 2], [7, 8]$	2
6	3	$[3, 3]$	$[1, 1], [3, 3], [7, 10]$	3
7	5	$[5, 5]$	$[5, 5], [7, 10]$	2
8	12	$[7, 12]$	$[7, 12]$	1
9	1	$[1, 1]$	$[1, 1], [7, 8]$	2
10	10	$[7, 10]$	$[7, 10]$	1
11	11	$[11, 11]$	$[7, 7], [9, 9], [11, 11]$	3
12	8	$[7, 8]$	$[7, 8]$	1
13	9	$[9, 9]$	$[7, 7], [9, 9]$	2
14	7	$[7, 7]$	$[7, 7]$	1
Total intervals				26

Tree 2. Post-order: 5, 2, 14, 12, 9, 6, 10, 7, 13, 11, 8, 3, 4, 1.

v	post	$I(v)$	Final $L(v)$	$ L(v) $
1	14	[1, 14]	[1, 14]	1
2	2	[1, 2]	[1, 2], [3, 6], [7, 7], [9, 9]	4
3	12	[3, 12]	[3, 12]	1
4	13	[13, 13]	[3, 4], [7, 7], [9, 11], [13, 13]	4
5	1	[1, 1]	[1, 1], [3, 5]	2
6	6	[3, 6]	[3, 6], [7, 7], [9, 9]	3
7	8	[7, 8]	[3, 4], [7, 8], [9, 9]	3
8	11	[9, 11]	[3, 4], [7, 7], [9, 11]	3
9	5	[3, 5]	[3, 5]	1
10	7	[7, 7]	[3, 4], [7, 7], [9, 9]	3
11	10	[9, 10]	[3, 3], [9, 10]	2
12	4	[3, 4]	[3, 4]	1
13	9	[9, 9]	[3, 3], [9, 9]	2
14	3	[3, 3]	[3, 3]	1
Total intervals				31

Which tree gives the better labelling?

	Tree 1	Tree 2
Total intervals (index size)	26	31
Largest single label	4 (vertex 3)	4 (vertices 2, 4)
Vertices with a single interval	6 (1, 4, 8, 10, 12, 14)	5 (1, 3, 9, 12, 14)
Label-size distribution (1/2/3/4)	6/5/2/1	5/3/4/2
Non-tree edges absorbed at no cost	2 (2 → 9, 4 → 13)	0

Why. A vertex's label starts as its own interval and then grows by inheriting intervals along its out-edges. Two distinct effects limit that growth.

- A non-tree edge $u \rightarrow w$ is *free* whenever w is already a tree descendant of u , because the tree path from u down to w carries $L(w)$ upward in any case. Note this is *not* a containment effect: in Tree 1, $L(9) = \{[1, 1], [7, 8]\}$ is not inside $I(2) = [1, 4]$, yet the edge $2 \rightarrow 9$ still costs nothing.
- Containment is the stronger effect, and the one that decides this comparison. If everything u can reach lies inside u 's own subtree, then every inherited interval has both endpoints among that subtree's post-numbers, so it is contained in $I(u)$ and deleted — and $L(u)$ collapses to the single interval $I(u)$. In both trees $L(u) = \{I(u)\}$ holds *exactly* for the vertices counted in the “single interval” row above.

So a spanning tree is good precisely when it keeps mutually reachable clusters together inside one subtree. Applying that to the two spanning trees:

- **Tree 1** puts the DAG's whole converging tail — 4, 8, 10, 11, 12, 13, 14 — into a single subtree rooted at 4 ($4 \rightarrow 8 \rightarrow \{10, 11\}$, $10 \rightarrow \{12, 13\}$, $12 \rightarrow 14$). Every DAG edge among those vertices therefore stays inside that subtree, so $L(4)$ collapses to the one interval $[7, 13]$, and likewise

$L(8) = [7, 12]$, $L(10) = [7, 10]$, $L(12) = [7, 8]$. On top of that, the non-tree edges $2 \rightarrow 9$ and $4 \rightarrow 13$ point back into the subtree they start from and cost nothing at all.

- **Tree 2** splits that same tail across three sibling branches of vertex 3 ($6 \rightarrow 9 \rightarrow 12 \rightarrow 14$, $7 \rightarrow 10$, $8 \rightarrow 11 \rightarrow 13$) and leaves vertex 4 as a *leaf*. A leaf's interval $I(4) = [13, 13]$ is a single point that can absorb nothing, so $4 \rightarrow 8$ and $4 \rightarrow 13$ push $L(4)$ up to four intervals. The cross-branch edges $6 \rightarrow 10$, $8 \rightarrow 10$, $10 \rightarrow 12$, $10 \rightarrow 13$ and $13 \rightarrow 14$ are likewise all non-tree edges between *different* branches, so their intervals propagate upward instead of being swallowed. That is why six of Tree 2's vertices need three or four intervals, against three in Tree 1.

Note that the *maximum* label size is 4 for both trees, so what separates them is the total index size — which is also what determines the space used and the cost of a query, since answering $u \rightsquigarrow v$ scans $L(u)$.

Answer. Tree 1 gives the better tree-cover labelling: 26 intervals against Tree 2's 31 (16% smaller), because Tree 1 keeps the DAG's converging tail $\{4, 8, 10, 11, 12, 13, 14\}$ inside one subtree, whereas Tree 2 scatters it over three sibling branches and leaves vertex 4 as an absorbing-nothing leaf.

Q4. One-layer GNN mean aggregation

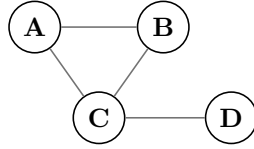


Figure 7: graph for Q4

$$E = \{A-B, A-C, B-C, C-D\}; \quad N(A) = \{B, C\}, \quad N(B) = \{A, C\}, \quad N(C) = \{A, B, D\}, \quad N(D) = \{C\}.$$

$$h_A^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad h_B^{(0)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h_C^{(0)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad h_D^{(0)} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad W = \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

(a) Vertices B and C

Vertex B, with $N(B) = \{A, C\}$:

$$h_{N(B)}^{(0)} = \frac{1}{2} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}, \quad \text{CONCAT}(h_B^{(0)}, h_{N(B)}^{(0)}) = (0, 1, 1, 0.5)^\top$$

$$W \cdot (0, 1, 1, 0.5)^\top = \begin{pmatrix} 1(0) - 1(1) + 1(1) + 0(0.5) \\ 0(0) + 1(1) - 1(1) + 1(0.5) \end{pmatrix} = \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} \xrightarrow{\text{ReLU}} h_B^{(1)} = \begin{pmatrix} 0.00 \\ 0.50 \end{pmatrix}$$

Vertex C, with $N(C) = \{A, B, D\}$:

$$h_{N(C)}^{(0)} = \frac{1}{3} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix}, \quad \text{CONCAT}(h_C^{(0)}, h_{N(C)}^{(0)}) = (1, 1, 1, 1/3)^\top$$

$$W \cdot (1, 1, 1, 1/3)^\top = \begin{pmatrix} 1 - 1 + 1 + 0 \\ 0 + 1 - 1 + 1/3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix} \xrightarrow{\text{ReLU}} h_C^{(1)} = \begin{pmatrix} 1.00 \\ 0.33 \end{pmatrix}$$

v	$N(v)$	Neighbour mean	Concatenated input	$h_v^{(1)}$
B	$\{A, C\}$	$(1, 0.5)^\top$	$(0, 1, 1, 0.5)^\top$	$(\mathbf{0.00}, \mathbf{0.50})^\top$
C	$\{A, B, D\}$	$(1, 1/3)^\top$	$(1, 1, 1, 1/3)^\top$	$(\mathbf{1.00}, \mathbf{0.33})^\top$

Answer. $h_B^{(1)} = (0.00, 0.50)^\top$ and $h_C^{(1)} = (1.00, 0.33)^\top$.

(b) After adding the edge $B-D$

Adding $B-D$ changes only $N(B)$ and $N(D)$, and no initial embedding changes, so A and C receive *identical* input vectors and cannot move. For B and D the input does change — but that on its own does not settle the question, since ReLU could clip two different pre-activations to the same value. Both are therefore recomputed explicitly:

v	$N(v)$ before $\rightarrow$ after	$h_v^{(1)}$ before	$h_v^{(1)}$ after	Changed?
A	$\{B, C\} \rightarrow \{B, C\}$	$(1.50, 0.50)^\top$	$(1.50, 0.50)^\top$	no
B	$\{A, C\} \rightarrow \{A, C, D\}$	$(0.00, 0.50)^\top$	$(\mathbf{0.33}, \mathbf{0.00})^\top$	yes
C	$\{A, B, D\} \rightarrow \{A, B, D\}$	$(1.00, 0.33)^\top$	$(1.00, 0.33)^\top$	no
D	$\{C\} \rightarrow \{B, C\}$	$(3.00, 0.00)^\top$	$(\mathbf{2.50}, \mathbf{0.50})^\top$	yes

The two recomputations, writing column vectors as transposed rows to keep them compact:

$$\begin{aligned}
B: \quad h_{N(B)}^{(0)} &= \frac{1}{3}[(1, 0) + (1, 1) + (2, 0)]^\top = \left(\frac{4}{3}, \frac{1}{3}\right)^\top \\
W \cdot (0, 1, \frac{4}{3}, \frac{1}{3})^\top &= (0 - 1 + \frac{4}{3} + 0, \quad 0 + 1 - \frac{4}{3} + \frac{1}{3})^\top = \left(\frac{1}{3}, 0\right)^\top \\
D: \quad h_{N(D)}^{(0)} &= \frac{1}{2}[(0, 1) + (1, 1)]^\top = (0.5, 1)^\top \\
W \cdot (2, 0, 0.5, 1)^\top &= (2 - 0 + 0.5 + 0, \quad 0 + 0 - 0.5 + 1)^\top = (2.5, 0.5)^\top
\end{aligned}$$

Both are already non-negative, so ReLU leaves them unchanged: $h_B^{(1)} = (0.33, 0.00)^\top$ (was $(0.00, 0.50)^\top$) and $h_D^{(1)} = (2.50, 0.50)^\top$ (was $(3.00, 0.00)^\top$). In fact ReLU never clips anything in this question — all eight pre-activations (four vertices, before and after) are already non-negative — so the masking it could in principle cause does not arise, and both endpoints genuinely move.

Answer. The affected vertices are $\{B, D\}$ — exactly the endpoints of the new edge. A and C are unchanged because a one-layer GNN only reads the 1-hop neighbourhood, and neither $N(A)$ nor $N(C)$ was touched by adding $B-D$.